Fidelity-Calibrated Distributionally Robust Model-Predictive Routing

A. Fidelity-Calibrated Distributionally Robust Model-Predictive Routing

To efficiently solve Problem $P1$, we propose a **Fidelity-Calibrated Distributionally Robust Model-Predictive Routing (FiDRO-MPC)** framework. Our approach separates network-state prediction from routing optimization while explicitly coupling them through a Digital-Twin fidelity measure.

FiDRO-MPC contains four interacting components: (i) topology-aware multi-step network-state prediction, (ii) Digital-Twin fidelity assessment and uncertainty calibration, (iii) counterfactual NDT evaluation of candidate routing decisions, and (iv) risk-sensitive receding-horizon traffic engineering.

The key principle is that a future network prediction should not have the same influence on a physical routing decision under all conditions. When the Digital Twin closely matches the physical network, the controller can confidently exploit predicted future states. Conversely, as synchronization error increases, the controller automatically enlarges its uncertainty set and adopts a more conservative routing policy.

B. Physical-State Acquisition and Twin Synchronization

At control epoch t, the SDN controller collects the physical telemetry vector

$$\mathbf{s}(t) = \{u_e(t), q_e(t), d_e(t), \ell_e(t), C_e(t)\}_{e\in\mathcal{E}}.$$

The Network Digital Twin maintains the corresponding state

$$\widetilde{\mathbf{s}}(t).$$

For link e, we compute the normalized twin discrepancy as

$$\delta_e(t) = \sum_{k=1}^{K} \omega_k \frac{|s_{e,k}(t) - \widetilde{s}_{e,k}(t)|}{R_k + \epsilon},$$

where R_k normalizes telemetry component k and

$$\sum_k \omega_k = 1.$$

We then define the link-level Digital-Twin fidelity as

$$\phi_e(t) = \exp[-\kappa\delta_e(t)].$$

Thus,

$$0 < \phi_e(t) \leq 1.$$

We obtain a network-level fidelity score as

$$\Phi(t) = \sum_{e \in \mathcal{E}} \nu_e(t) \phi_e(t),$$

where

$$\sum_e \nu_e(t) = 1.$$

The weights $\nu_e(t)$ may depend on the traffic carried by link e, thereby assigning greater importance to synchronization errors on highly utilized links.

C. Topology-Aware Multi-Step Prediction

We employ a graph-based predictor because network traffic evolution depends jointly on temporal correlations and dependencies between neighboring links.

We recommend an **MPNN-GRU predictor**.

We represent the physical network as the graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}).$$

For link e, the initial embedding is

$$\mathbf{h}_e^{(0)}(t) = \mathbf{W}_x \mathbf{s}_e(t).$$

We propagate spatial information through message passing as

$$\mathbf{m}_e^{(l)} = \sum_{j \in \mathcal{N}(e)} \alpha_{ej}^{(l)} \mathbf{W}_m^{(l)} \mathbf{h}_j^{(l)},$$

and

$$\mathbf{h}_e^{(l+1)} = \sigma\left(\mathbf{W}_s^{(l)} \mathbf{h}_e^{(l)} + \mathbf{m}_e^{(l)}\right).$$

We capture temporal evolution using

$$\mathbf{g}_e(t) = \mathrm{GRU}\left(\mathbf{h}_e(t), \mathbf{g}_e(t-1)\right).$$

The decoder predicts the network variables over horizon

$$h = 1, \ldots, H.$$

Instead of producing only a point estimate, the predictor generates

$$\widehat{\mu}_{e,h}, \qquad \widehat{q}_{e,h}^{0.05}, \qquad \widehat{q}_{e,h}^{0.95}.$$

The predicted variables can include

$$\widehat{\lambda}_{f,h}, \quad \widehat{C}_{e,h}, \quad \widehat{d}_{e,h}, \quad \widehat{\ell}_{e,h}.$$

We train the model using a combination of point-prediction and quantile losses:

$$\mathcal{L}_{\mathrm{pred}} = \lambda_M \mathcal{L}_{\mathrm{MSE}} + \lambda_Q \sum_{\tau \in \{0.05, 0.5, 0.95\}} \mathcal{L}_{\mathrm{pinball}}^{\tau}.$$

We do not claim novelty for the graph predictor itself. Its role is to provide a topology-aware probabilistic forecasting module for the optimization framework.

D. Conformal Calibration of Prediction Uncertainty

Prediction intervals produced directly by a neural network may be miscalibrated. FiDRO-MPC therefore applies split conformal calibration using a validation dataset.

For horizon h, we define the non-conformity score

$$r_{i,h} = \max\{\widehat{q}^{L}_{i,h} - y_{i,h}, y_{i,h} - \widehat{q}^{U}_{i,h}, 0\}.$$

We compute the conformal correction as

$$q^{\text{conf}}_{h} = \text{Quantile}_{1-\alpha_c}\{r_{1,h}, \ldots, r_{N_c,h}\}.$$

The calibrated interval becomes

$$[\widehat{q}^{L}_{e,h} - q^{\text{conf}}_{h},\ \widehat{q}^{U}_{e,h} + q^{\text{conf}}_{h}].$$

FiDRO-MPC further modifies this uncertainty according to Digital-Twin fidelity.

For link e,

$$r^{\text{FiDRO}}_{e,h} = r^{0}_{e,h}\left[1 + \beta_{\phi}(1 - \phi_e(t))\right]. \tag{18}$$

Therefore,

$$\frac{\partial r^{\text{FiDRO}}_{e,h}}{\partial \phi_e} = -\beta_{\phi} r^{0}_{e,h} < 0.$$

Hence, uncertainty automatically increases as twin fidelity deteriorates.

This constitutes an important difference from conventional prediction-assisted routing, where the optimizer usually treats forecasts independently of the current reliability of the virtual network representation.

E. Fidelity-Calibrated Ambiguity Set

Let

$$\widehat{\mathbb{P}}_t = \frac{1}{N_s}\sum_{s=1}^{N_s} \delta_{\widehat{\boldsymbol{\xi}}^{s}_{t}}$$

denote the empirical distribution constructed using prediction-error samples.

FiDRO-MPC defines a Wasserstein ambiguity set

$$\mathcal{P}_t = \left\{\mathbb{Q} : W(\mathbb{Q}, \widehat{\mathbb{P}}_t) \le \epsilon_t\right\}. \tag{19}$$

We dynamically adjust the ambiguity radius according to

$$\epsilon_t = \epsilon_0 + \beta_{\phi}(1 - \Phi(t)) + \beta_e \bar{e}(t), \tag{20}$$

where $\bar{e}(t)$ denotes the recent rolling prediction error.

Equation (20) provides the central coupling between the Digital Twin and the routing optimizer.

For a highly accurate twin,

$$\Phi(t) \to 1$$

and low forecast error,

$$\bar{e}(t) \to 0,$$

we obtain

$$\epsilon_t \approx \epsilon_0.$$

If the physical network and its twin become poorly synchronized,

$$\Phi(t) \downarrow,$$

then

$$\epsilon_t \uparrow .$$

The controller therefore protects itself against model mismatch by optimizing over a larger set of plausible future distributions.

F. Candidate-Path Generation

Solving the routing problem over every possible simple path is computationally prohibitive. FiDRO-MPC therefore maintains a candidate path set

$$\mathcal{P}_f^K$$

for each flow f.

The initial path set contains:

$$K_1$$

shortest paths according to propagation delay,

$$K_2$$

least-utilized paths, and, when available,

$$K_3$$

link-disjoint or partially disjoint paths.

Thus,

$$|\mathcal{P}_f^K| \ll |\mathcal{P}_f|.$$

Unlike static K-shortest-path routing, we do not use path costs directly for the final decision. The paths merely define the feasible action set supplied to the counterfactual NDT and the predictive optimizer.

G. Counterfactual Digital-Twin Evaluation

A major feature of FiDRO-MPC is that future network conditions are conditioned on the candidate routing action.

Let

$$\mathbf{x}_h$$

denote a candidate routing configuration for future control epoch $t + h$.

For uncertainty realization $\boldsymbol{\xi}_h^\omega$, the Digital Twin evaluates

$$\widehat{\mathbf{s}}_{t+h+1}^\omega = \mathcal{M}_{\mathrm{DT}} \left(\widehat{\mathbf{s}}_{t+h}^\omega, \mathbf{x}_h, \boldsymbol{\xi}_h^\omega \right). \tag{21}$$

Consequently,

$$\widehat{\rho}^{\omega}_{e,h} = \frac{\sum_f \sum_{p \in \mathcal{P}_f^K} A_{ep} \widehat{\lambda}^{\omega}_{f,h} x_{fph}}{\widehat{C}^{\omega}_{e,h}}. \tag{22}$$

This avoids the conceptual error of forecasting congestion first and then assuming that the forecast remains unchanged after traffic is rerouted.

Instead, FiDRO-MPC estimates

$$P(\mathbf{s}_{t+h} \mid \mathbf{s}_t, \mathbf{x}_{1:h}),$$

rather than merely

$$P(\mathbf{s}_{t+h} \mid \mathbf{s}_t).$$

Hence, routing decisions and future network states are explicitly coupled.

H. Risk Function

For scenario ω, we define the congestion-risk term

$$G^{\omega}_h = \sum_{e \in \mathcal{E}} \left[\rho^{\omega}_{e,h} - \rho^{\text{safe}}_e \right]^+.$$

The SLA-risk component is

$$S^{\omega}_h = \sum_f \left[\frac{D^{\omega}_{f,h} - D^{\max}_f}{D^{\max}_f} \right]^+ + \zeta_L \sum_f \left[\frac{L^{\omega}_{f,h} - L^{\max}_f}{L^{\max}_f} \right]^+.$$

The combined violation loss is

$$V^{\omega}_h = \omega_G G^{\omega}_h + \omega_S S^{\omega}_h. \tag{23}$$

FiDRO-MPC controls the upper tail of V_h using

$$\text{CVaR}_{\alpha}(V_h) = \min_{\eta_h} \left\{ \eta_h + \frac{1}{1-\alpha} \mathbb{E}[(V_h - \eta_h)^+] \right\}. \tag{24}$$

Typical experiments should evaluate several values such as

$$\alpha \in \{0.90, 0.95, 0.99\}.$$

I. Distributionally Robust Receding-Horizon Problem

For a planning horizon H, FiDRO-MPC solves

$$\begin{aligned} \min_{\mathbf{x},\mathbf{z},\boldsymbol{\eta}} \quad & \sum_{h=1}^{H} \gamma^{h-1} \Big[J^{\text{nom}}_h + \lambda_R \sup_{\mathbb{Q} \in \mathcal{P}_t} \text{CVaR}^{\mathbb{Q}}_{\alpha}(V_h) \Big] \\ & + \lambda_C \sum_{h=1}^{H} \sum_f c^R_f z_{fh}, \end{aligned} \tag{25}$$

where

$$J_h^{\text{nom}} = w_D \mathbb{E}[D_h] + w_L \mathbb{E}[L_h] + w_U \mathbb{E}[U_h].$$

The optimization is subject to path-selection, flow-conservation, present-state capacity, SLA, route-update-budget, and dwell-time constraints.

The term

$$\sup_{\mathbb{Q} \in \mathcal{P}_t} \text{CVaR}_\alpha^{\mathbb{Q}}(V_h)$$

is critical.

Conventional stochastic routing asks:

What is the expected performance under the estimated distribution?

FiDRO-MPC instead asks:

What tail risk can arise under any plausible distribution close to the estimated one?

J. Fidelity-Adaptive Risk Weighting

We may introduce an additional mechanism to make the risk sensitivity itself depend on twin confidence:

$$\lambda_R(t) = \lambda_R^0 + \chi[1 - \Phi(t)]. \tag{26}$$

Consequently,

$$\Phi(t) \downarrow \quad \Rightarrow \quad \lambda_R(t) \uparrow .$$

Equations (20) and (26) operate differently.

Equation (20) enlarges the set of plausible future distributions, whereas Eq. (26) increases the penalty associated with poor tail outcomes.

For the initial paper, I recommend using Eq. (20) as the primary mechanism and treating Eq. (26) as an ablation or optional extension to prevent unnecessary parameterization.

K. Decomposition-Based Solution

Directly solving (25) for all future scenarios and candidate routing configurations is expensive.

FiDRO-MPC therefore adopts an **anytime column-and-constraint generation procedure**.

Let

$$\Omega^{(r)}$$

denote the set of adverse scenarios included at iteration r.

The restricted master problem solves

$$\min_{\mathbf{x}, \mathbf{z}, \eta} J(\mathbf{x}, \mathbf{z}, \eta)$$

considering only

$$\omega \in \Omega^{(r)}.$$

This yields

$$\mathbf{x}^{(r)}, \mathbf{z}^{(r)}.$$

For this routing configuration, an adversarial subproblem determines the future network realization producing the greatest congestion/SLA exposure within the current ambiguity set:

$$\boldsymbol{\xi}^* = \arg\max_{\boldsymbol{\xi} \in \mathcal{U}_t} V(\mathbf{x}^{(r)}, \boldsymbol{\xi}). \tag{27}$$

If the corresponding robust-risk violation satisfies

$$\Delta^{(r)} \leq \varepsilon_{\mathrm{CG}},$$

the procedure terminates.

Otherwise,

$$\Omega^{(r+1)} = \Omega^{(r)} \cup \{\boldsymbol{\xi}^*\},$$

and we solve the master problem again.

Therefore, computational effort is concentrated on future scenarios that can actually invalidate or significantly degrade the current routing solution.

L. Receding-Horizon Execution

Assume FiDRO-MPC determines

$$\mathbf{x}_1^*, \mathbf{x}_2^*, \ldots, \mathbf{x}_H^*.$$

We install only

$$\mathbf{x}_1^*$$

in the physical SDN network.

We discard the remaining decisions.

At control epoch $t+1$, we collect new telemetry and repeat the entire process.

Hence,

$$\mathbf{s}(t+1) \to \Phi(t+1) \to \mathcal{P}_{t+1} \to \text{new optimization}.$$

This receding-horizon policy prevents long-range prediction errors from permanently determining physical-network behavior.

M. Proposed Algorithm

Algorithm 1: FiDRO-MPC

Input: Physical topology $\mathcal{G}$, previous route configuration $\mathbf{x}(t-1)$, candidate-path sets $\{\mathcal{P}_f^K\}$, prediction horizon H, CVaR confidence α, telemetry window W, reconfiguration budget B^{route}.

Output: Routing decision $\mathbf{x}(t)$.

1. Obtain network telemetry $\mathbf{s}(t)$ from the SDN controller.
2. Synchronize the Network Digital Twin and obtain $\widetilde{\mathbf{s}}(t)$.
3. Compute link-level fidelity scores $\phi_e(t)$ and global fidelity $\Phi(t)$.
4. Use the MPNN-GRU predictor to estimate multi-step traffic and network states for $h = 1, \ldots, H$.

5. Apply conformal calibration to the predictive intervals.
6. Construct the empirical forecast-error distribution $\widehat{\mathbb{P}}_t$.
7. Compute the fidelity-aware Wasserstein radius

$$\epsilon_t = \epsilon_0 + \beta_\phi(1 - \Phi(t)) + \beta_e \bar{e}(t).$$

8. Construct ambiguity set $\mathcal{P}_t$.
9. Initialize the adversarial scenario set $\Omega^{(0)}$ using nominal and high-load scenarios.
10. Warm-start the routing variables using the currently deployed configuration.
11. Solve the restricted DR-CVaR master problem.
12. Evaluate the resulting routes using counterfactual NDT rollouts.
13. Solve the adversarial scenario-generation subproblem.
14. If the maximum robust-risk violation is larger than $\varepsilon_{\mathrm{CG}}$, add the adverse scenario and return to Step 11.
15. Otherwise, obtain the horizon solution

$$\{\mathbf{x}_1^*, \ldots, \mathbf{x}_H^*\}.$$

16. Apply only $\mathbf{x}_1^*$ through the SDN controller.
17. Observe the physical-network response and update the prediction-error history and twin fidelity.
18. Proceed to the next control epoch.

1 Baselines + Exact solution

In addition to the above method: You need to implement the following 4 base-lines and 1 exact method.
Then alltogether 6 algorithms you should compare. You will have 12 plots (plotting the significant parameters) and compare these 6 algorithms against them.
BASELINES:1) Reactive Multi-Objective Traffic Engineering (There is no Digital Twin prediction, no future horizon, and no uncertainty modeling.)

2) Deterministic Digital-Twin MPC: Use exactly the same prediction model and planning horizon as FiDRO-MPC, but optimize against the mean prediction (The 2025 IEEE work on NDT route optimization already demonstrates GNN-based what-if route evaluation, so a point-predictive NDT baseline is well motivated.)

3) Graph-PPO Predictive Routing :Use a graph encoder for network state representation followed by PPO

eXACT METHOD: EXACT-MILP: For small networks, construct an arc-based deterministic equivalent of the complete scenario problem. which can be solved using Gurobi or CPLEX branch-and-bound/branch-and-cut.

2)